It's not uncommon to see a class that assembles its functionality out of modules, but ActiveRecord::Base does this on a large scale. The code above does nothing but extend or include tens of modules. (Plus one additional line, the call to run_load_hooks, that allows some of those modules to run their own configuration code after they've been autoloaded.) As it turns out, many of the modules included by Base also include even more modules.

This is where the autoloading mechanism pays off. ActiveRecord::Base doesn't need to require a module's source code and then include the module. Instead, it just includes the module. Thanks to autoloading, classes such as Base can do lots of module inclusions with minimal code.

In some cases, it's not too hard to find which module a specific method in Base comes from. For example, persistence methods such as save come from ActiveRecord::Persistence:

gems/activerecord-4.1.0/lib/active_record/persistence.rb
```
module ActiveRecord
  module Persistence
    def save(*)  # ...
    def save!(*) # ...
    def delete   # ...
```

Other method definitions are harder to find. In *A Short Active Record Example*, on page 171, you looked at validation methods such as valid? and validate. Let's go hunting for them.

The Validations Modules

Among the other modules, ActiveRecord::Base includes a module named ActiveRecord::Validations. This module looks like a good candidate to define methods such as valid? and validate. Indeed, if you look in ActiveRecord::Validations, you'll find the definition of valid?—but no validate:

gems/activerecord-4.1.0/lib/active_record/validations.rb
```
module ActiveRecord
  module Validations
    include ActiveModel::Validations
    # ...
    def valid?(context = nil) # ...
```

Where is validate? We can look for the answer in ActiveModel::Validations, a module that ActiveRecord::Validation includes. This module comes from Active Model, a library that Active Record depends on. Sure enough, if you look into its source, you'll find that validate is defined in ActiveModel::Validation.